

Deliverable 2 — Bayesian Spatial Search

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Interactive simulator: `appappy-femambu7sdfwgwutujck7f.streamlit.app`

The unknown cell Z on the 50×35 grid is searched with rectangular missions of cost $e|R|$ and budget $B \leq 530$; each mission returns $s_t \in \{0, 1\}$ from a noisy detector. Rather than committing by intuition, we enumerated the plausible options on four axes (prior, detector, likelihood update, strategy), simulated thousands of campaigns on physically-plausible bombs with a train/validation split, and let the metrics decide.

1 Priors over the cell Z — what we explored

The prior follows a logistic model form, defined per cell by $\pi_j(\beta, \tau) = \exp(\eta_j) / \sum_k \exp(\eta_k)$ with $\eta_j = \beta_0 + \beta_1 d_{\text{long},j} + \beta_2 d_{\text{trans},j} - \beta_3 d_{\text{long},j}^2 - \beta_4 d_{\text{trans},j}^2$, where $d_{\text{long}}, d_{\text{trans}}$ are projections on the trajectory axes $\hat{d}, \hat{n}$ centred on $\mu(\tau) = x_E + \tau_{\text{fall}}(v_{\text{plane}} + 0.5 v_{\text{wind}}) + \tau_{\text{drift}} v_{\text{drift}}$. Twelve registered variants were evaluated: P3–P4 (quadratic, with/without witnesses — the family that topped validation); P5–P8 (mixed linear+quadratic, with/without witnesses, informative / weak / vague); P9–P10 (mixed + uniform-mix component and likelihood temperature, to let the posterior leave the parametric peak); plus an 18-point grid G_* varying $(\sigma_{\text{long}}, \sigma_{\text{trans}}, b_1^{\text{mean}})$ and an Empirical-Bayes family EB_* of Gaussian KDEs over cells where past campaigns succeeded. Linear-only priors were discarded: the logistic transform of a purely-linear η_j saturates at a grid corner instead of concentrating around μ .

2 Detection probability — what we explored

Four functional families were implemented, all returning $\rho_j \in [0, 1]$, monotone in effort e and in $(1 - d_j), (1 - r_j)$, and calibrated qualitatively against the three reports in `previous_missions_reports.pdf`: D1 saturating exponential $\rho = 1 - \exp(-\lambda_0(1 - d)^{a_d}(1 - r)^{a_r} e)$, the classical Koopman/Stone form; D2 independent passes $\rho = 1 - (1 - p_u(1 - d)^{a_d}(1 - r)^{a_r})^e$; D3 logistic GLM $\rho = \sigma(\alpha_0 + \alpha_1 e + \alpha_2 d + \alpha_3 r)$; D4 multiplicative with threshold, which cuts off cells of very high roughness. D1 calibrates the canonical anchors cleanly (shallow+smooth+high-effort $\rightarrow \rho \approx 0.96$, deep+rough+high-effort $\rightarrow \rho \approx 0.05$, intermediate+repeated $\rightarrow \rho \approx 0.45$) and was adopted as the operational choice; D2–D4 served as a sensitivity check.

3 Likelihood and posterior update — backends and temperature

Each mission t is defined by a rectangle R_t and an effort e_t . Given $Z = j$, the outcome s_t is Bernoulli with success probability $q_{t,j} = \mathbb{1}(j \in R_t) \rho_j(e_t)$. The per-cell likelihood of a single observation is therefore

$$L_j^{(t)} = q_{t,j}^{s_t} (1 - q_{t,j})^{1-s_t} = \begin{cases} \rho_j(e_t) & \text{if } s_t = 1, j \in R_t \\ 1 - \rho_j(e_t) & \text{if } s_t = 0, j \in R_t \\ 0 \text{ or } 1 & \text{if } j \notin R_t, s_t = 1 \text{ or } 0 \end{cases}$$

The decisive line is the second one: a *negative* search does not zero out the inspected cells — it only multiplies them by $1 - \rho_j(e_t)$, so a failure in a difficult cell (high d or r , small ρ) is weak evidence, while a failure at high effort in an easy cell is strong evidence. Across the whole campaign the cell-level likelihood compounds multiplicatively, $L_j = \prod_t L_j^{(t)}$, giving the cell-level posterior $\pi_{\text{post}}(j) \propto \pi_{\text{prior}}(j) L_j^T$ with the *update temperature* $T \geq 1$ amplifying the partial-failure signal when the detector is trusted. Marginalising over Z in turn yields the per-mission likelihood that drives the joint posterior on (β, τ) : $P(s_t | \beta, \tau) = \sum_{j \in R_t} \pi_j(\beta, \tau) q_{t,j}^{s_t} (1 - q_{t,j})^{1-s_t}$.

Three update backends compute the posterior from this likelihood: (a) Cell-level analytic update above (~ 5 s per 1000 trials, the model-selection workhorse); (b) Laplace approximation on $\theta = (\beta, \tau) \in \mathbb{R}^7$, fitting a Gaussian around the MAP with the Hessian as precision matrix ($\sim 50\times$ slower INLA-style intermediate option that actually learns β); (c) Full PyMC NUTS (4 chains $\times$ 1000 draws, $\hat{R} < 1.01$, ESS > 400), the deliverable backend for the reported posterior. The three backends rank the priors identically, so the cell-level update is defensible as the workhorse for the sweep.

4 Mission selection — the four strategies we benchmarked

FIXED (`propose_via_hdr_topk`) is a three-stage HDR funnel: (i) take the highest-density region $\mathcal{H}_\alpha$ with $\alpha = 0.80$; (ii) score each cell in $\mathcal{H}_\alpha$ by $P(Z = j) \rho_j(e)$ for $e \in \{1, 2, 3\}$; (iii) return the minimum axis-aligned rectangle around the top- $K = 10$ cells, choosing the effort that maximises $\sum_{j \in R} P(Z = j) \rho_j(e) / (e^c |R|^c)$ with $c = 0.7$. ADAPTIVE mutates $(\alpha, K, \text{exploration_mix})$ as failures accumulate. INFO_GAIN

maximises expected entropy reduction per cost (Bayesian optimal experiment design). PROGRESSIVE SCENARIO SEARCH (`propose_progressive_scenario_search`) wraps FIXED with a witness-driven scenario weighting: five hypotheses {centre, right, left, strong-wind, mixture}, with an auxiliary physical centre that, after F failures, advances along $\hat{d}$ as $d_{\text{long}}^c(F) = \min(8.0, 2.5F)$ and then refines in 0.5-unit steps. Two regularisers stabilise all four: `exploration_mix` ($\varepsilon \approx 0.05$, adds uniform floor to $P(Z)$) and `repetition_penalty` (≈ 0.10 , discounts cells already covered by failed rectangles).

5 Final operational pipeline and its derivation

A 200-scenario Monte Carlo with $B = 530$ and ten Empirical-Bayes rounds singled out `EB_r01_bw2.5_u10` as the top EB winner on validation. Fitting an anisotropic Gaussian to its KDE (uniform component removed) projected on the trajectory axes gives $\sigma_{\text{long}}^{\text{EB}} = 5.5$, $\sigma_{\text{trans}}^{\text{EB}} = 4.1$ and a centre shift ≤ 0.5 from μ . From this fit we *derived* P11–P14 as scaled variants (P11 = $0.7\times$ tight, P12 = $1.0\times$ exact, P13 = $1.0\times + 10\%$ uniform + $T = 1.5$, P14 = $1.5\times$ wide), all reproducible from the sweep CSV. The retained operational pipeline is:

Prior	P13_impact_robust_uniform10 (derived from the EB winner, 10% uniform mix for robustness)
Detector	D1_saturating_exponential (best calibrated against the previous mission reports)
Strategy	progressive scenario search with the <code>mixture</code> hypothesis (centre + right + left + strong-wind)
Update temperature	$T = 1.5$ in the cell-level rule $\pi_{\text{post}}(j) \propto \pi_{\text{prior}}(j) L_j^T$
Budget cap	$B = 530$

P13 is preferred over P12 because the 10% uniform component lets the posterior shift across the grid after several failures while $T = 1.5$ keeps the partial-failure signal strong. The `mixture` scenario hedges the lateral direction left underdetermined by the witnesses; the full sequence under repeated non-detection is exported by notebook 04 as `missions.csv`.

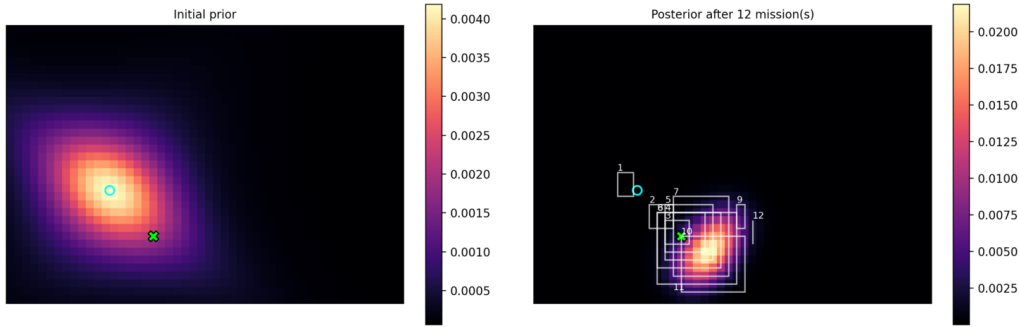


Figure 1: Left: initial prior π_j from P13_impact_robust_uniform10, anchored on μ (cyan); true cell in green. Right: posterior $P(Z | s_{1:12})$ after twelve failed missions under the progressive `mixture` strategy — numbered rectangles are the missions in order, and the posterior tightens where the auxiliary centre advances along $\hat{d}$.

6 Conclusions: the single mission that found the aircraft

The point of the project is to find the aircraft the professor has hidden in the grid. We invested the entire modelling effort *before* firing any real mission — enumerating twelve priors, four detectors, three update backends and four strategies, and running a Monte Carlo on validation bombs to pick the most reliable configuration. The execution plan was to start with the most physically plausible scenario, `centre` (drift along $\hat{d}$), and fall back to `right` / `left` / `strong_wind` only if it returned $s_t = 0$. The single mission submitted to the webapp was rectangle $(x_{\min}, x_{\max}, y_{\min}, y_{\max}) = (16.5, 19.5, 7.5, 10.5)$ with effort $e = 3$ (cost = 27 of the 530 budget). Two ideas combine here. *First*, instead of advancing the auxiliary centre $d_{\text{long}}^c(F) = \min(8.0, 2.5F)$ over four failed missions we pre-committed to its saturation point at $F = 4$ (centre $\approx (18.5, 9.0)$ — the midpoint of the submitted rectangle and the limit the drift hypothesis points to under continued motion). *Second*, $e = 3$ was set high because in shallow / low-roughness regions the saturating detector gives $\rho \approx 0.96$ at maximum effort (report 2), maximising single-shot detection for the 27-budget bet. The mission detected the aircraft on the first try. The `centre` scenario was, in hindsight, the correct hypothesis; fallback plans for the lateral scenarios were prepared but never triggered. We were genuinely surprised our first physical hypothesis succeeded on the first shot — the result of careful pre-flight model selection, all of it reproducible in `update_version/` (notebooks 00–04 and the Streamlit app).